

UNIT - I Introduction

Concept of Operating Systems- Generations of Operating systems-Types of Operating Systems-OS Services-System Calls-Structure of an OS - Layered, Monolithic, Microkernel Operating Systems-Concept of Virtual Machine-Case study on UNIX and WINDOWS Operating System.

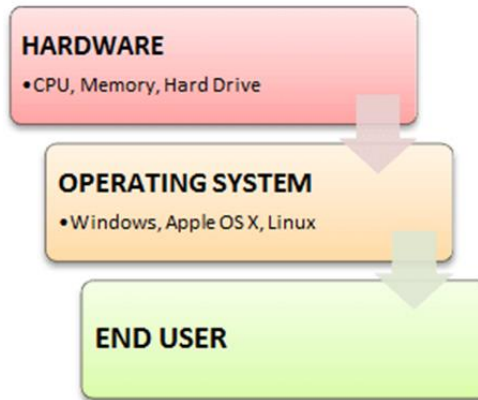
Concept of Operating Systems:

- An operating system acts as an intermediary between the user of a computer and computer hardware.
- It is a software that manages the computer hardware.
- The purpose of an operating system is to provide an environment in which a user can execute programs in a convenient and efficient manner.
- The hardware must provide appropriate mechanisms to ensure the correct operation of the computer system and to prevent user programs from interfering with the proper operation of the system.
- Every computer system must have at least one operating system to run other programs. Applications like Browsers, MS Office, Notepad Games, etc., need some environment to run and perform its tasks.
- A more common definition is that the **operating system is the one program running at all times on the computer (usually called the kernel), with all else being application programs.**
- An operating system is concerned with the allocation of resources and services, such as memory, processors, devices, and information.
- The operating system correspondingly includes programs to manage these resources, such as a traffic controller, a scheduler, memory management module, I/O programs, and a file system.
- It is not possible for the user to use any computer or mobile device without having an operating system.

Functions of Operating system – Operating system performs three functions:

1. **Convenience:** An OS makes a computer more convenient to use.
2. **Efficiency:** An OS allows the computer system resources to be used in an efficient manner.

3. **Ability to Evolve:** An OS should be constructed in such a way as to permit the effective development, testing and introduction of new system functions at the same time without interfering with service.



Operating system as User Interface –

Every general-purpose computer consists of the hardware, operating system, system programs, and application programs. The hardware consists of memory, CPU, ALU, and I/O devices, peripheral device, and storage device. System program consists of compilers, loaders, editors, OS, etc. The application program consists of business programs, database programs.

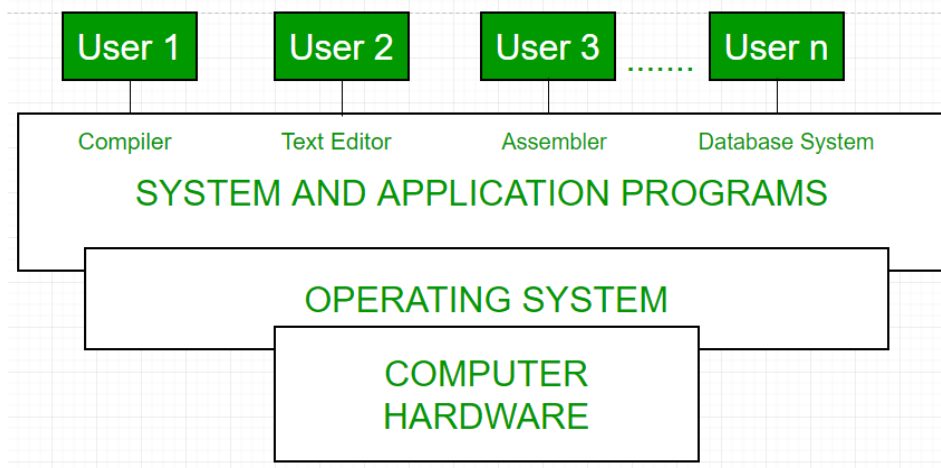


Fig1: Conceptual view of a computer system

The operating system is a set of special programs that run on a computer system that allows it to work properly. It performs basic tasks such as recognizing input from the keyboard, keeping track of files and directories on the disk, sending output to the display screen and controlling peripheral devices.

OS is designed to serve two basic purposes:

1. It controls the allocation and use of the computing System's resources among the various user and tasks.
2. It provides an interface between the computer hardware and the programmer that simplifies and makes feasible for coding, creation, debugging of application programs.

The Operating system must support the following tasks. The task are:

1. Provides the facilities to create, modification of programs and data files using an editor.
2. Access to the compiler for translating the user program from high level language to machine language.
3. Provide a loader program to move the compiled program code to the computer's memory for execution.
4. Provide routines that handle the details of I/O programming.

History of OS (Generations of Operating systems):

A computer is an electronic device that manipulates information or data. It has the ability to store, retrieve, and process data.

Nowadays, a computer can be used to type documents, send email, play games, and browse the Web. It can also be used to edit or create spreadsheets, presentations, and even videos. But the evolution of this complex system started around 1940 with the first Generation of Computer and evolving ever since.

There are five generations of computers.

1. First Generation

- ***Introduction:***

1. 1946-1959 is the period of first generation computer.
2. J.P.Eckert and J.W.Mauchy invented the first successful electronic computer called ENIAC, ENIAC stands for "Electronic Numeric Integrated And Calculator".

- ***Few Examples are:***

1. ENIAC
2. EDVAC
3. UNIVAC
4. IBM-701
5. IBM-650

- ***Advantages:***

1. It made use of vacuum tubes which are the only electronic component available during those days.
2. These computers could calculate in milliseconds.

- ***Disadvantages:***

1. These were very big in size, weight was about 30 tones.
2. These computers were based on vacuum tubes.
3. These computers were very costly.
4. It could store only a small amount of information due to the presence of magnetic drums.
5. As the invention of first generation computers involves vacuum tubes, so another disadvantage of these computers was, vacuum tubes require a large cooling system.
6. Very less work efficiency.
7. Limited programming capabilities and punch cards were used to take inputs.
8. Large amount of energy consumption.
9. Not reliable and constant maintenance is required.

2. Second Generation

- ***Introduction:***

1. 1959-1965 is the period of second-generation computer.
2. 3. Second generation computers were based on Transistor instead of vacuum tubes.

- ***Few Examples are:***

1. Honeywell 400
2. IBM 7094
3. CDC 1604
4. CDC 3600
5. UNIVAC 1108

- ***Advantages:***

1. Due to the presence of transistors instead of vacuum tubes, the size of electron component decreased. This resulted in reducing the size of a computer as compared to first generation computers.
2. Less energy and not produce as much heat as the first generation.
3. Assembly language and punch cards were used for input.
4. Low cost than first generation computers.
5. Better speed, calculate data in microseconds.
6. Better portability as compared to first generation

- ***Disadvantages:***

1. A cooling system was required.
2. Constant maintenance was required.
3. Only used for specific purposes.

3. Third Generation

- ***Introduction:***

1. 1965-1971 is the period of third generation computer.
2. These computers were based on Integrated circuits.
3. IC was invented by Robert Noyce and Jack Kilby In 1958-1959.
4. IC was a single component containing number of transistors.

- ***Few Examples are:***

1. PDP-8
2. PDP-11
3. ICL 2900
4. IBM 360
5. IBM 370

- ***Advantages:***

1. These computers were cheaper as compared to second-generation computers.
2. They were fast and reliable.
3. Use of IC in the computer provides the small size of the computer.
4. IC not only reduce the size of the computer but it also improves the performance of the computer as compared to previous computers.
5. This generation of computers has big storage capacity.
6. Instead of punch cards, mouse and keyboard are used for input.
7. They used an operating system for better resource management and used the concept of time-sharing and multiple programming.

8. These computers reduce the computational time from microseconds to nanoseconds.

- ***Disadvantages:***

1. IC chips are difficult to maintain.
2. The highly sophisticated technology required for the manufacturing of IC chips.
3. Air conditioning is required.

4. Fourth Generation

- ***Introduction:***

1. 1971-1980 is the period of fourth generation computer.
2. This technology is based on Microprocessor.
3. A microprocessor is used in a computer for any logical and arithmetic function to be performed in any program.
4. Graphics User Interface (GUI) technology was exploited to offer more comfort to users.

- ***Few Examples are:***

1. IBM 4341
2. DEC 10
3. STAR 1000
4. PUP 11

- ***Advantages:***

1. Fastest in computation and size get reduced as compared to the previous generation of computer.
2. Heat generated is negligible.
3. Small in size as compared to previous generation computers.
4. Less maintenance is required.

5. All types of high-level language can be used in this type of computers.

- ***Disadvantages:***

1. The Microprocessor design and fabrication are very complex.
2. Air conditioning is required in many cases due to the presence of ICs.
3. Advance technology is required to make the ICs.

5. Fifth Generation

- ***Introduction:***

1. The period of the fifth generation in 1980-onwards.
2. This generation is based on artificial intelligence.
3. The aim of the fifth generation is to make a device which could respond to natural language input and are capable of learning and self-organization.
4. This generation is based on ULSI(Ultra Large Scale Integration) technology resulting in the production of microprocessor chips having ten million electronic component.

- ***Few Examples are:***

1. Desktop
2. Laptop
3. NoteBook
4. UltraBook
5. Chromebook

- ***Advantages:***

1. It is more reliable and works faster.
2. It is available in different sizes and unique features.
3. It provides computers with more user-friendly interfaces with multimedia features.

- **Disadvantages:**

1. They need very low-level languages.
2. They may make the human brains dull and doomed.

GENERATION DEVICE	YEAR	ELECTRONIC DEVICE USED	TYPES OF OS
First	1945-55	Vaccum Tubes	Plug Boards
Second	1955-65	Transistors	Batch Systems
Third	1965-80	Integrated Circuits(IC)	Multiprogramming
Fourth	Since 1980	Large Scale Integration	Personal Computers

Summery:

- Operating systems were first developed in the late 1950s to manage tape storage
- The General Motors Research Lab implemented the first OS in the early 1950s for their IBM 701
- In the mid-1960s, operating systems started to use disks
- In the late 1960s, the first version of the Unix OS was developed
- The first OS built by Microsoft was DOS. It was built in 1981 by purchasing the 86-DOS software from a Seattle company
- The present-day popular OS Windows first came to existence in 1985 when a GUI was created and paired with MS-DOS.

Types of Operating Systems

An Operating System performs all the basic tasks like managing file, process, and memory. Thus operating system acts as manager of all the resources, i.e. **resource manager**. Thus operating system becomes an interface between user and machine.

Types of Operating Systems: Some of the widely used operating systems are as follows-

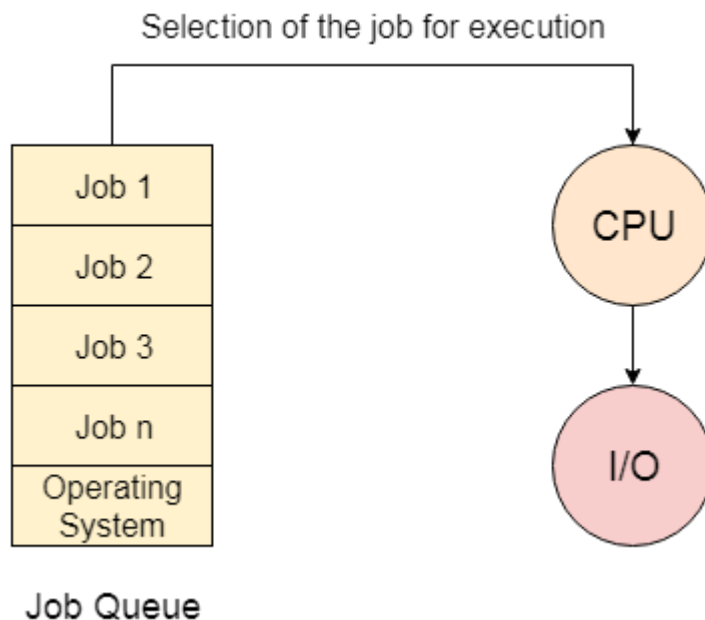
1. Batch Operating System

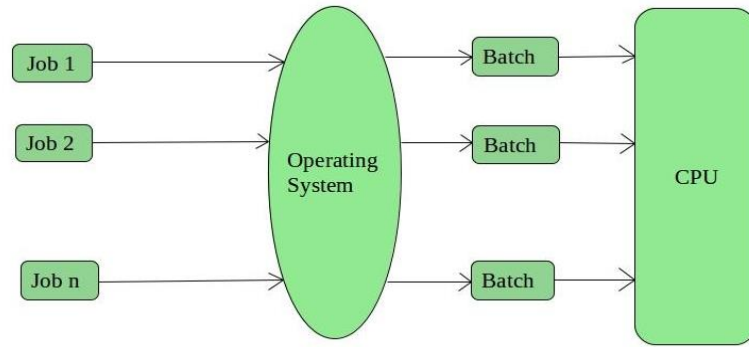
In the era of 1970s, the Batch processing was very popular. The Jobs were executed in batches. People were used to have a single computer which was called mainframe.

In Batch operating system, access is given to more than one person; they submit their respective jobs to the system for the execution.

The system put all of the jobs in a queue on the basis of first come first serve and then executes the jobs one by one. The users collect their respective output when all the jobs get executed.

This type of operating system does not interact with the computer directly. There is an operator which takes similar jobs having same requirement and group them into batches. It is the responsibility of operator to sort the jobs with similar needs.





Advantages of Batch Operating System:

- It is very difficult to guess or know the time required by any job to complete. Processors of the batch systems know how long the job would be when it is in queue
- Multiple users can share the batch systems
- The idle time for batch system is very less
- It is easy to manage large work repeatedly in batch systems

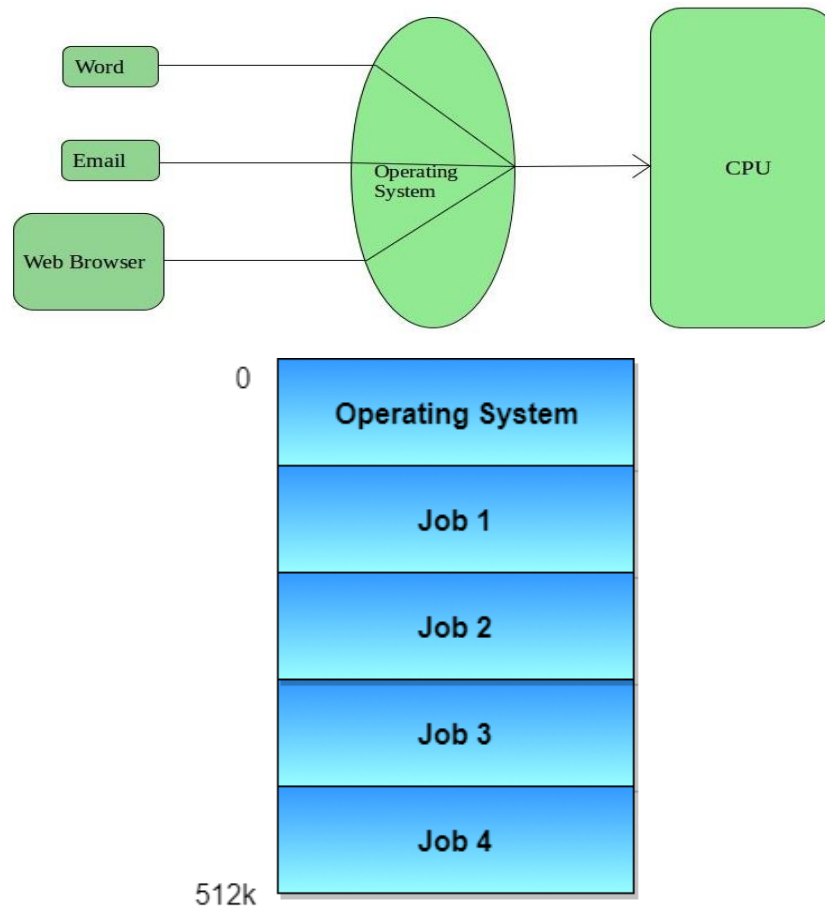
Disadvantages of Batch Operating System:

- **Starvation:** Batch processing suffers from starvation. If there are five jobs J1, J2, J3, J4, J4 and J5 present in the batch. If the execution time of J1 is very high then other four jobs will never be going to get executed or they will have to wait for a very high time. Hence the other processes get starved.
- **Not Interactive:** Batch Processing is not suitable for the jobs which are dependent on the user's input. If a job requires the input of two numbers from the console then it will never be going to get it in the batch processing scenario since the user is not present at the time of execution.
- The computer operators should be well known with batch systems
- Batch systems are hard to debug
- It is sometime costly
- The other jobs will have to wait for an unknown time if any job fails

Examples of Batch based Operating System: Payroll System, Bank Statements etc.

2. Multi-Tasking/Time-Sharing Operating Systems:

Each task is given some time to execute, so that all the tasks work smoothly. Each user gets time of CPU as they use single system. These systems are also known as **Multitasking Systems**. The task can be from single user or from different users also. The time that each task gets to execute is called quantum. After this time interval is over OS switches over to next task.



Advantages of Time-Sharing OS:

- Each task gets an equal opportunity
- Less chances of duplication of software
- CPU idle time can be reduced

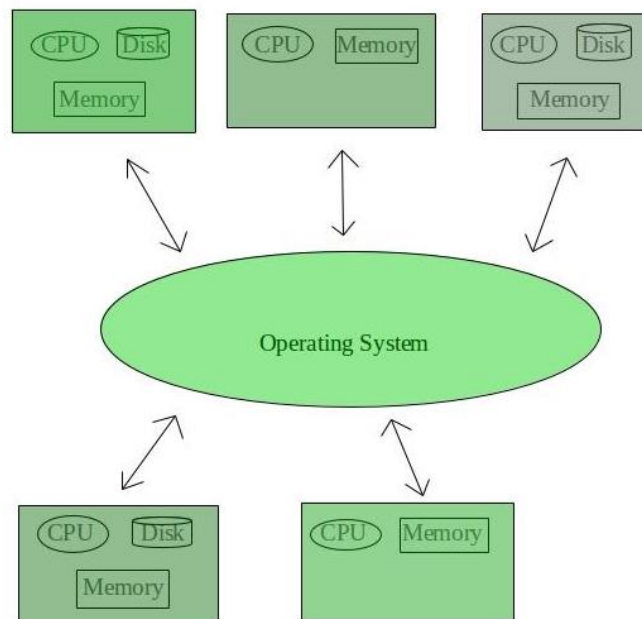
Disadvantages of Time-Sharing OS:

- Reliability problem
- One must have to take care of security and integrity of user programs and data
- Data communication problem

Examples of Time-Sharing OSs are: Multics, Unix etc.

3. Distributed Operating System:

These types of operating system is a recent advancement in the world of computer technology and are being widely accepted all-over the world. Various autonomous interconnected computers communicate each other using a shared communication network. Independent systems possess their own memory unit and CPU. These are referred as **loosely coupled systems** or distributed systems. These system's processors differ in size and function. The major benefit of working with these types of operating system is that it is always possible that one user can access the files or software which are not actually present on his system but on some other system connected within this network i.e., remote access is enabled within the devices connected in that network.



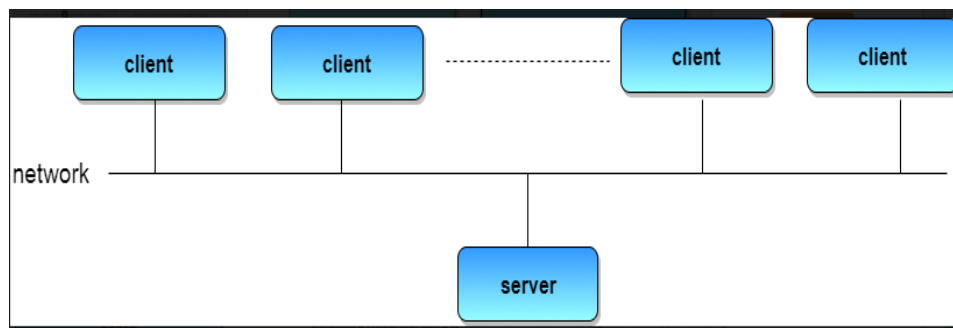
Types of Distributed Operating Systems

Following are the two types of distributed operating systems used:

1. Client-Server Systems
2. Peer-to-Peer Systems

1. Client-Server Systems

Centralized systems today act as **server systems** to satisfy requests generated by **client systems**. The general structure of a client-server system is depicted in the figure below:



Server Systems can be broadly categorized as: **Compute Servers** and **File Servers**.

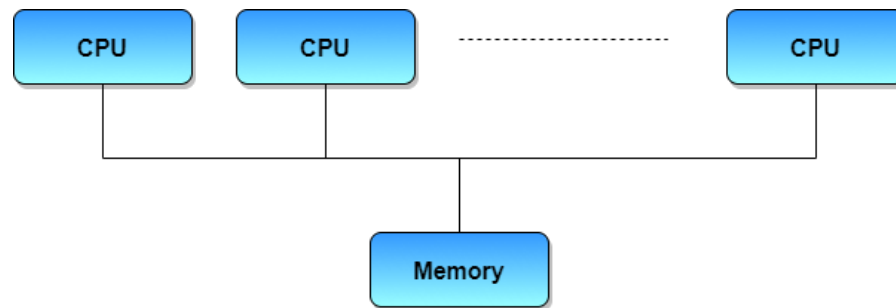
- **Compute Server systems**, provide an interface to which clients can send requests to perform an action, in response to which they execute the action and send back results to the client.
- **File Server systems**, provide a file-system interface where clients can create, update, read, and delete files.

2. Peer-to-Peer Systems

The growth of computer networks - especially the Internet and World Wide Web (WWW) – has had a profound influence on the recent development of operating systems. When PCs were introduced in the 1970s, they were designed for **personal** use and were generally considered standalone computers. With the beginning of widespread public use of the Internet in the 1990s for electronic mail and FTP, many PCs became connected to computer networks.

In contrast to the **Tightly Coupled** systems, the computer networks used in these applications consist of a collection of processors that do not share memory or a

clock. Instead, each processor has its own local memory. The processors communicate with one another through various communication lines, such as high-speed buses or telephone lines. These systems are usually referred to as loosely coupled systems (or distributed systems). The general structure of a client-server system is depicted in the figure below:



Advantages of Distributed Operating System:

- Failure of one will not affect the other network communication, as all systems are independent from each other
- Electronic mail increases the data exchange speed
- Since resources are being shared, computation is highly fast and durable
- Load on host computer reduces
- These systems are easily scalable as many systems can be easily added to the network
- Delay in data processing reduces

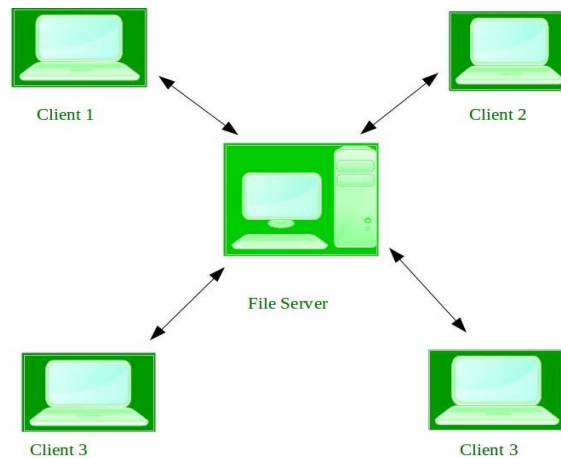
Disadvantages of Distributed Operating System:

- Failure of the main network will stop the entire communication
- To establish distributed systems the language which are used are not well defined yet
- These types of systems are not readily available as they are very expensive. Not only that the underlying software is highly complex and not understood well yet

Examples of Distributed Operating System are- LOCUS etc.

4. Network Operating System:

These systems run on a server and provide the capability to manage data, users, groups, security, applications, and other networking functions. These type of operating systems allow shared access of files, printers, security, applications, and other networking functions over a small private network. One more important aspect of Network Operating Systems is that all the users are well aware of the underlying configuration, of all other users within the network, their individual connections etc. and that's why these computers are popularly known as **tightly coupled systems**.



Advantages of Network Operating System:

- Highly stable centralized servers
- Security concerns are handled through servers
- New technologies and hardware up-gradation are easily integrated to the system
- Server access are possible remotely from different locations and types of systems

Disadvantages of Network Operating System:

- Servers are costly
- User has to depend on central location for most operations
- Maintenance and updates are required regularly

Examples of Network Operating System are: Microsoft Windows Server 2003, Microsoft Windows Server 2008, UNIX, Linux, Mac OS X, Novell NetWare, and BSD etc.

There are many types of operating system exists in the current scenario:

5. Real-Time Operating System:

These types of OSs serves the real-time systems. The time interval required to process and respond to inputs is very small. This time interval is called **response time**. In Real Time systems, each job carries a certain deadline within which the Job is supposed to be completed, otherwise the huge loss will be there or even if the result is produced then it will be completely useless.

Real-time systems are used when there are time requirements are very strict like missile systems (if you want to drop a missile then the missile is supposed to be dropped with certain precision), air traffic control systems, robots etc.

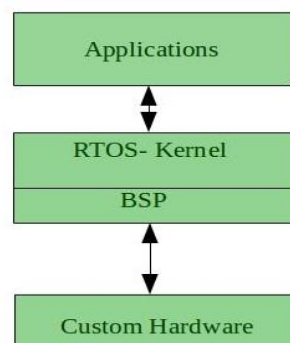
Two types of Real-Time Operating System which are as follows:

- **Hard Real-Time Systems:**

These OSs are meant for the applications where time constraints are very strict and even the shortest possible delay is not acceptable. These systems are built for saving life like **automatic parachutes or air bags** which are required to be readily available in case of any accident. Virtual memory is almost never found in these systems.

- **Soft Real-Time Systems:**

These OSs are for applications where for time-constraint is less strict.



The BSP provides a common application programming interface (API) and a stable environment for the real-time operating system.

Advantages of RTOS:

- **Maximum Consumption:** Maximum utilization of devices and system, thus more output from all the resources
- **Task Shifting:** Time assigned for shifting tasks in these systems are very less. For example in older systems it takes about 10 micro seconds in shifting one task to another and in latest systems it takes 3 micro seconds.
- **Focus on Application:** Focus on running applications and less importance to applications which are in queue.
- **Real time operating system in embedded system:** Since size of programs are small, RTOS can also be used in embedded systems like in transport and others.
- **Error Free:** These types of systems are error free.
- **Memory Allocation:** Memory allocation is best managed in these type of systems.

Disadvantages of RTOS:

- **Limited Tasks:** Very few tasks run at the same time and their concentration is very less on few applications to avoid errors.
- **Use heavy system resources:** Sometimes the system resources are not so good and they are expensive as well.
- **Complex Algorithms:** The algorithms are very complex and difficult for the designer to write on.
- **Device driver and interrupt signals:** It needs specific device drivers and interrupt signals to response earliest to interrupts.
- **Thread Priority:** It is not good to set thread priority as these systems are very less prone to switching tasks.

Examples of Real-Time Operating Systems are: Scientific experiments, medical imaging systems, industrial control systems, weapon systems, robots, air traffic control systems, etc.

6. Mobile OS

Mobile operating systems are those OS which is especially that are designed to power smartphones, tablets, and wearables devices. Some most famous mobile operating systems are Android and iOS, but others include BlackBerry, Web, and watchOS.

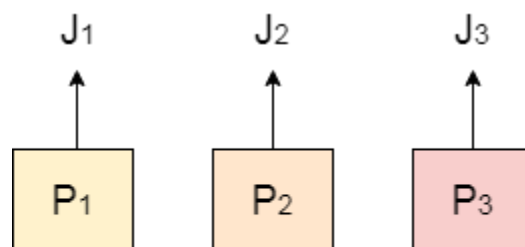
7. Multiprogramming Operating System

Multiprogramming is an extension to the batch processing where the CPU is kept always busy. Each process needs two types of system time: CPU time and IO time.

In multiprogramming environment, for the time a process does its I/O, The CPU can start the execution of other processes. Therefore, multiprogramming improves the efficiency of the system.

8. Multiprocessing Operating System

In Multiprocessing, Parallel computing is achieved. It consists of several processors that share a common physical memory. in the system which can execute more than one process at the same time. This will increase the throughput of the system. This system provides higher computing power and speed. In multiprocessor system all processors operate under single operating system. Multiplicity of the processors and how they do act together are transparent to the others.



Multi Processing

Advantages of Multiprocessor Systems

1. Enhanced performance
2. Execution of several tasks by different processors concurrently, increases the system's throughput without speeding up the execution of a single task.

3. If possible, system divides task into many subtasks and then these subtasks can be executed in parallel in different processors. Thereby speeding up the execution of single tasks.

9. Desktop Systems

Earlier, CPUs and PCs lacked the features needed to protect an operating system from user programs. PC operating systems therefore were neither **multiuser** nor **multitasking**. However, the goals of these operating systems have changed with time; instead of maximizing CPU and peripheral utilization, the systems opt for maximizing user convenience and responsiveness. These systems are called **Desktop Systems** and include PCs running Microsoft Windows and the Apple Macintosh. Operating systems for these computers have benefited in several ways from the development of operating systems for **mainframes**.

Microcomputers were immediately able to adopt some of the technology developed for larger operating systems. On the other hand, the hardware costs for microcomputers are sufficiently **low** that individuals have sole use of the computer, and CPU utilization is no longer a prime concern. Thus, some of the design decisions made in operating systems for mainframes may not be appropriate for smaller systems.

10. Clustered Systems

- Like parallel systems, clustered systems gather together multiple CPUs to accomplish computational work.
- Clustered systems differ from parallel systems, however, in that they are composed of two or more individual systems coupled together.
- The definition of the term clustered is **not concrete**; the general accepted definition is that clustered computers share storage and are closely linked via LAN networking.
- Clustering is usually performed to provide **high availability**.
- A layer of cluster software runs on the cluster nodes. Each node can monitor one or more of the others. If the monitored machine fails, the monitoring machine can take ownership of its storage, and restart the application(s) that were running on the failed machine. The failed machine can remain down,

but the users and clients of the application would only see a brief interruption of service.

- **Asymmetric Clustering** - In this, one machine is in hot standby mode while the other is running the applications. The hot standby host (machine) does nothing but monitor the active server. If that server fails, the hot standby host becomes the active server.
- **Symmetric Clustering** - In this, two or more hosts are running applications, and they are monitoring each other. This mode is obviously more efficient, as it uses all of the available hardware.
- **Parallel Clustering** - Parallel clusters allow multiple hosts to access the same data on the shared storage. Because most operating systems lack support for this simultaneous data access by multiple hosts, parallel clusters are usually accomplished by special versions of software and special releases of applications.

Clustered technology is rapidly changing. Clustered system's usage and its features should expand greatly as **Storage Area Networks(SANs)**. SANs allow easy attachment of multiple hosts to multiple storage units. Current clusters are usually limited to two or four hosts due to the complexity of connecting the hosts to shared storage.

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11. Handheld Systems

Handheld systems include **Personal Digital Assistants(PDAs)**, such as Palm-Pilots or Cellular Telephones with connectivity to a network such as the Internet. They are usually of limited size due to which most handheld devices have a small amount of memory, include slow processors, and feature small display screens.

- Many handheld devices have between **512 KB** and **8 MB** of memory. As a result, the operating system and applications must manage memory

efficiently. This includes returning all allocated memory back to the memory manager once the memory is no longer being used.

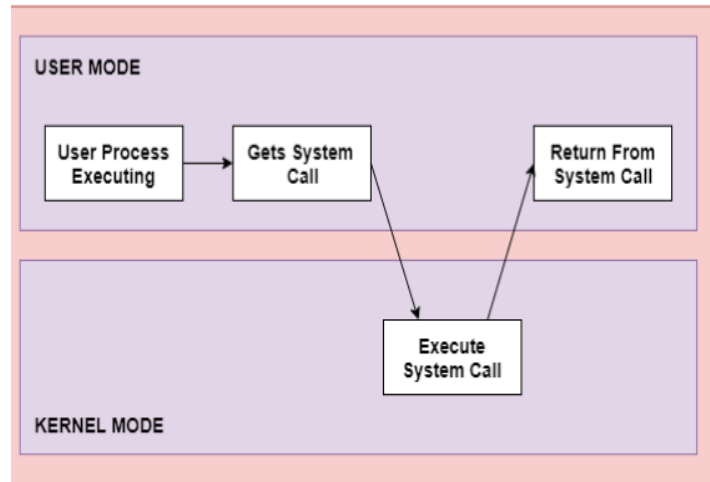
- Currently, many handheld devices do **not use virtual memory** techniques, thus forcing program developers to work within the confines of limited physical memory.
- Processors for most handheld devices often run at a fraction of the speed of a processor in a PC. Faster processors require **more power**. To include a faster processor in a handheld device would require a **larger battery** that would have to be replaced more frequently.
- The last issue confronting program designers for handheld devices is the small display screens typically available. One approach for displaying the content in web pages is **web clipping**, where only a small subset of a web page is delivered and displayed on the handheld device.

Some handheld devices may use wireless technology such as **BlueTooth**, allowing remote access to e-mail and web browsing. **Cellular telephones** with connectivity to the Internet fall into this category. Their use continues to expand as network connections become more available and other options such as cameras and MP3 players, expand their utility.

System Call

In computing, a **system call** is the programmatic way in which a computer program requests a service from the kernel of the operating system it is executed on. A system call is a way for programs to **interact with the operating system**. A computer program makes a system call when it makes a request to the operating system's kernel. System call **provides** the services of the operating system to the user programs via Application Program Interface (API). It provides an interface between a process and operating system to allow user-level processes to request services of the operating system. System calls are the only entry points into the kernel system. All programs needing resources must use system calls.

A figure representing the execution of the system call is given as follows



As can be seen from this diagram, the processes execute normally in the user mode until a system call interrupts this. Then the system call is executed on a priority basis in the kernel mode. After the execution of the system call, the control returns to the user mode and execution of user processes can be resumed.

Services Provided by System Calls:

An Operating System provides services to both the users and to the programs.

- It provides programs an environment to execute.
- It provides users the services to execute the programs in a convenient manner.

Following are a few common services provided by an operating system –

- a. Program execution
- b. I/O operations
- c. File System manipulation
- d. Communication
- e. Error Detection
- f. Resource Allocation
- g. Protection

a. Program execution

Operating systems handle many kinds of activities from user programs to system programs like printer spooler, name servers, file server, etc. Each of these activities is encapsulated as a process.

A process includes the complete execution context (code to execute, data to manipulate, registers, OS resources in use). Following are the major activities of an operating system with respect to program management –

- Loads a program into memory.
- Executes the program.
- Handles program's execution.
- Provides a mechanism for process synchronization.
- Provides a mechanism for process communication.
- Provides a mechanism for deadlock handling.

b. I/O Operation

An I/O subsystem comprises of I/O devices and their corresponding driver software. Drivers hide the peculiarities of specific hardware devices from the users.

An Operating System manages the communication between user and device drivers.

- I/O operation means read or write operation with any file or any specific I/O device.
- Operating system provides the access to the required I/O device when required.

c. File system manipulation

A file represents a collection of related information. Computers can store files on the disk (secondary storage), for long-term storage purpose.

Examples of storage media include magnetic tape, magnetic disk and optical disk drives like CD, DVD. Each of these media has its own properties like speed, capacity, data transfer rate and data access methods.

A file system is normally organized into directories for easy navigation and usage. These directories may contain files and other directions. Following are the major activities of an operating system with respect to file management –

- Program needs to read a file or write a file.
- The operating system gives the permission to the program for operation on file.
- Permission varies from read-only, read-write, denied and so on.
- Operating System provides an interface to the user to create/delete files.
- Operating System provides an interface to the user to create/delete directories.
- Operating System provides an interface to create the backup of file system.

d. Communication

In case of distributed systems which are a collection of processors that do not share memory, peripheral devices, or a clock, the operating system manages communications between all the processes. Multiple processes communicate with one another through communication lines in the network.

The OS handles routing and connection strategies, and the problems of contention and security. Following are the major activities of an operating system with respect to communication –

- Two processes often require data to be transferred between them
- Both the processes can be on one computer or on different computers, but are connected through a computer network.
- Communication may be implemented by two methods, either by Shared Memory or by Message Passing.

e. Error handling

Errors can occur anytime and anywhere. An error may occur in CPU, in I/O devices or in the memory hardware. Following are the major activities of an operating system with respect to error handling –

- The OS constantly checks for possible errors.
- The OS takes an appropriate action to ensure correct and consistent computing.

f. Resource Management

In case of multi-user or multi-tasking environment, resources such as main memory, CPU cycles and files storage are to be allocated to each user or job. Following are the major activities of an operating system with respect to resource management –

- The OS manages all kinds of resources using schedulers.
- CPU scheduling algorithms are used for better utilization of CPU.

g. Protection

Considering a computer system having multiple users and concurrent execution of multiple processes, the various processes must be protected from each other's activities.

Protection refers to a mechanism or a way to control the access of programs, processes, or users to the resources defined by a computer system. Following are the major activities of an operating system with respect to protection –

- The OS ensures that all access to system resources is controlled.
- The OS ensures that external I/O devices are protected from invalid access attempts.
- The OS provides authentication features for each user by means of passwords.

Types of System Calls: There are 5 different categories of system calls –

1. **Process control:** end, abort, create, terminate, allocate and free memory.
2. **File management:** create, open, close, delete, read file etc.

3. **Device management:** reading from device buffers, writing into device buffers etc.
4. **Information maintenance:** handle information and its transfer between the operating system and the user program.
5. **Communication:** useful for interprocess communication and also deal with creating and deleting a communication connection.

Examples of Windows and Unix System Calls –

	WINDOWS	UNIX
Process Control	CreateProcess() ExitProcess() WaitForSingleObject()	fork() exit() wait()
File Manipulation	CreateFile() ReadFile() WriteFile() CloseHandle()	open() read() write() close()
Device Manipulation	SetConsoleMode() ReadConsole() WriteConsole()	ioctl() read() write()
Information Maintenance	GetCurrentProcessID() SetTimer() Sleep()	getpid() alarm() sleep()
Communication	CreatePipe() CreateFileMapping() MapViewOfFile()	pipe() shmget() mmap()
Protection	SetFileSecurity() InitializeSecurityDescriptor() SetSecurityDescriptorGroup()	chmod() umask() chown()

There are many different system calls as shown above. Details of some of those system calls are as follows –

open()

The `open()` system call is used to provide access to a file in a file system. This system call allocates resources to the file and provides a handle that the process uses to refer to the file. A file can be opened by multiple processes at the same time or be restricted to one process. It all depends on the file organisation and file system.

read()

The `read()` system call is used to access data from a file that is stored in the file system. The file to read can be identified by its file descriptor and it should be opened using `open()` before it can be read. In general, the `read()` system call takes three arguments i.e. the file descriptor, buffer which stores read data and number of bytes to be read from the file.

write()

The `write()` system call writes the data from a user buffer into a device such as a file. This system call is one of the ways to output data from a program. In general, the `write` system call takes three arguments i.e. file descriptor, pointer to the buffer where data is stored and number of bytes to write from the buffer.

close()

The `close()` system call is used to terminate access to a file system. Using this system call means that the file is no longer required by the program and so the buffers are flushed, the file metadata is updated and the file resources are de-allocated.

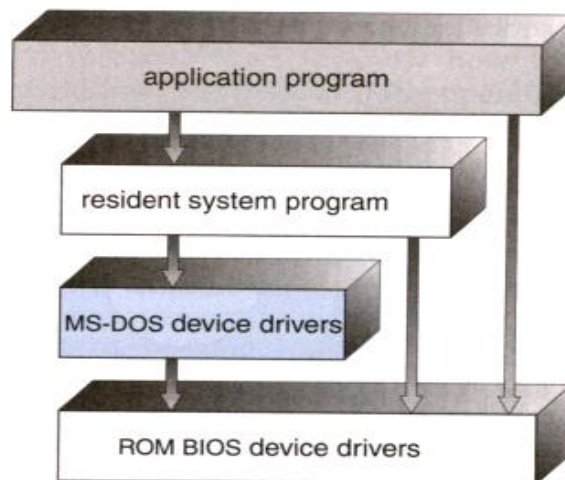
Structure of an OS:

A structure of an Operating System determines how it has been designed and how it functions. There are numerous ways of designing a new structure of an Operating system. Here, we will learn about six combinations that have been tested and tried. These six combinations are **monolithic systems, layered systems, microkernels, client-server models, virtual machines, and exokernels.**

Note: Before we get started it's important to understand what a kernel is.

- When your computer is running in kernel mode, all the permissions are available (think of it as an administrator).
- In macOS, this is known as giving 'root' access. In Windows, you invoke this by running applications as 'administrator.'
- If not the kernel mode, the computer normally runs in user mode, where a lot of permissions are not given to the user that may damage the system files.

Simple Structure

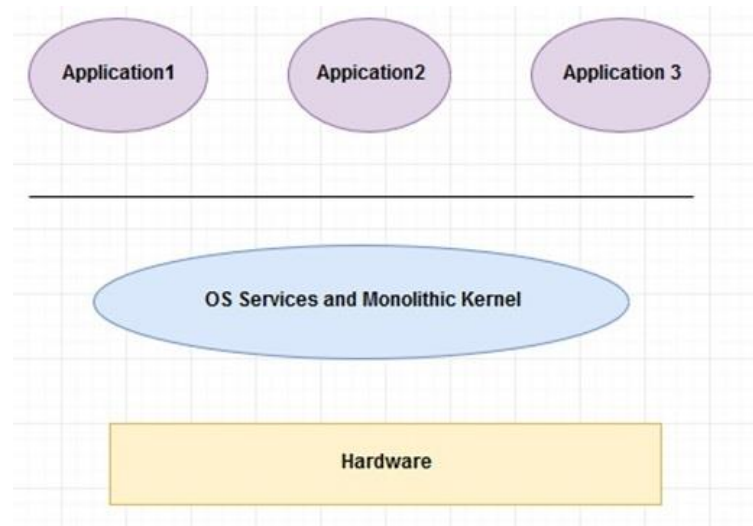


In MS-DOS, applications may bypass the operating system.

- Operating systems such as MS-DOS and the original UNIX did not have well-defined structures.
- There was no CPU Execution Mode (user and kernel), and so errors in applications could cause the whole system to crash.

I. Monolithic System structure in an Operating System

In monolithic kernel based systems, each application has its own address space. Therefore, each application is secure. Also, the kernel contains all the OS services. Therefore, the applications can request services from the kernel. Some OS services are file system, CPU Scheduler, network access, memory manager etc. However, the OS is in a separate address space. Therefore, it is secure from the normal applications and malfunctioned applications.



If an application requires a service, the hardware address space of the application switches to the hardware address space of the operating system to execute it.

In this organizational structure, the entire operating system runs as a single program in the kernel mode. An operating system is a collection of various procedures linked together in a binary file. In this system, any procedure can call any other procedure. Since it is running in kernel mode itself, it has all the permissions to call whatever it wants.

In terms of information hiding, there is none. All procedures are running in kernel mode, so they have access to all modules and packages of other procedures.

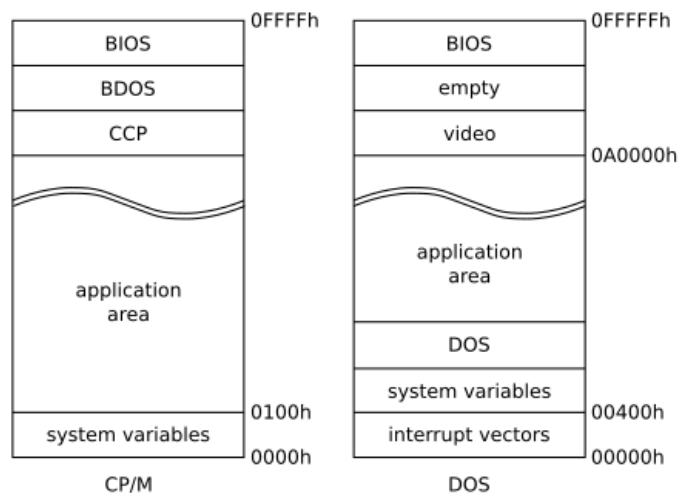
However, using this approach without any restrictions can lead to thousands of procedure calls, and this can lead to a messy system. For this purpose, the actual OS is constructed in a hierarchy. All the individual procedures are compiled into a single executable file using the system linker.

Even a monolithic system has a structure in which it can run in user mode. There already is a basic structure given by the organization

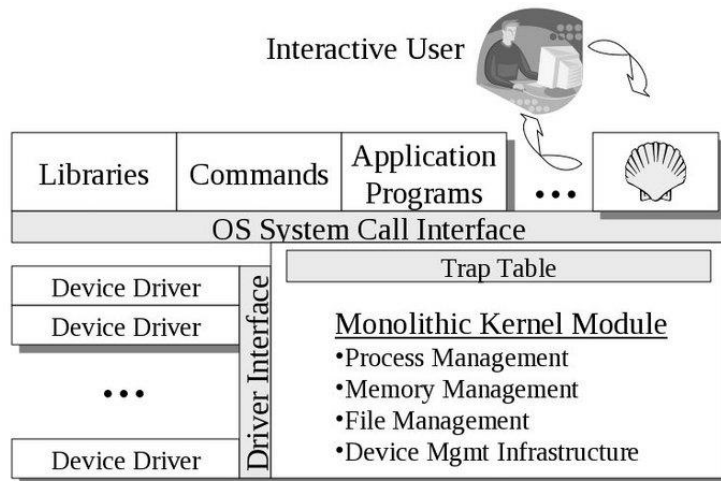
1. The main procedure that invokes the requested service procedures.
2. A set of service procedures that carry out system calls.
3. A set of utility procedures that help out the system procedures.

Simple Monolithic Operating Systems Example

CP/M and DOS are simple examples of monolithic operating systems. Both CP/M and DOS are operating systems that share a single address space with the applications. In CP/M, the 16 bit address space starts with system variables and the application area and ends with three parts of the operating system, namely CCP (Console Command Processor), BDOS (Basic Disk Operating System) and BIOS (Basic Input/Output System). In DOS, the 20 bit address space starts with the array of interrupt vectors and the system variables, followed by the resident part of DOS and the application area and ending with a memory block used by the video card and BIOS.

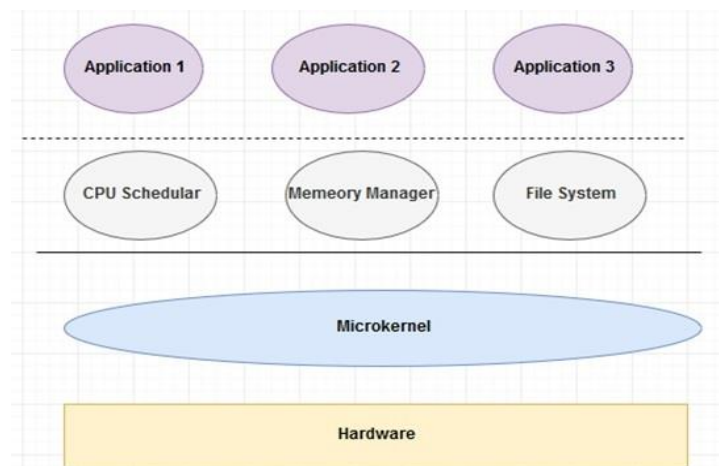


Most contemporary operating systems, including Linux and Windows, are also considered monolithic, even though their structure is certainly significantly different from the simple examples of CP/M and DOS.



II. Microkernels system in an operating system

Microkernel is a type of kernel that allows customization of the operating system. It runs on privileged mode and provides low-level address space management and Inter Process Communication (IPC). Moreover, OS services such as file system, virtual memory manager, and CPU scheduler are on top of the microkernel. Each service has its own address space to make them secure. Besides, the applications also have their own address spaces. Therefore, there is protection among applications, OS Services and kernel.



When the application requests the OS services for a service, the OS services communicate with each other to provide the required service to the application. Here, the Inter Process Communication (IPC) helps to establish this communication. Overall, microkernel-based OS provides a great level of

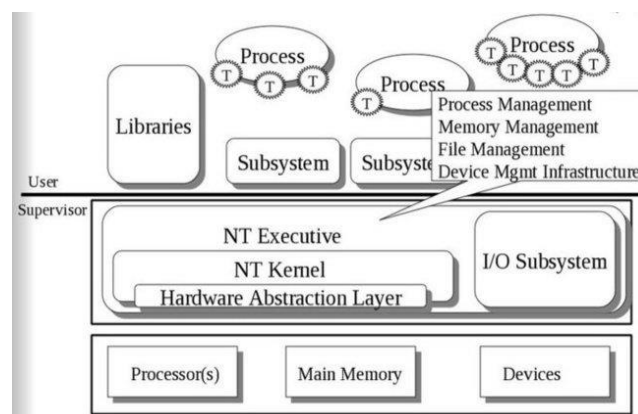
extensibility. It is also possible to customize the services of the operating system depending on the requirements of the application.

Traditionally, all the layers of the OS in a layered system went into the kernel. So they all had root access to the OS, and any small bug in any layer could be fatal to the OS. Famous examples of a microkernel system include Integrity, K42, PikeOS, Symbian, and MINIX 3.

The primary purpose of this system is to provide high reliability. Because of the high reliability that it provides, the applications of microkernels can be seen in real-time, industrial, avionics (electronics fitted in aircraft and aviation), and military applications that are mission-critical and require high reliability.

Working

- The operating system is split into small, well-defined modules, of which only one, the microkernel, runs in kernel mode. The rest of the modules run as powerless ordinary user processes.
- Running each device driver and file system as separate user processes is a fail-safe method as a bug in one of the drivers, will fail only that component.
- A bug easily references an invalid memory address and brings the system to a grinding halt instantly.



A Microkernel architecture

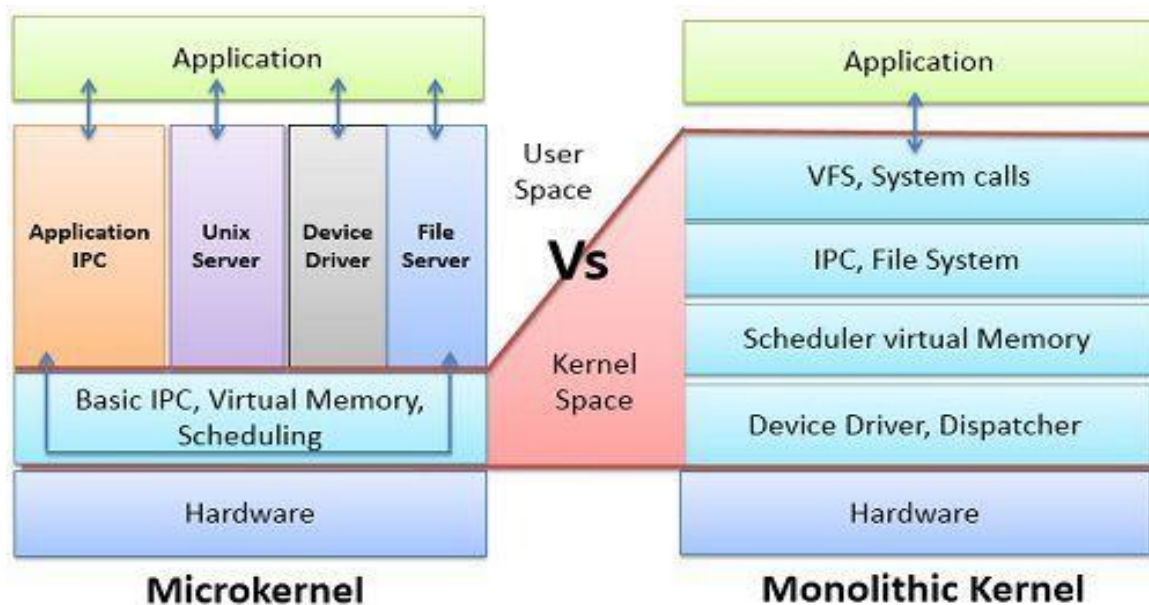
Example: MINIX 3

- ✓ The MINIX 3 is an OS written in C with 3200 lines of code and about 800 lines of code for the assembler to handle low-level functions like catching interrupts or switching processes.
- ✓ The C code handles managing and scheduling processes, handles interprocess communication.
- ✓ It also provides a set of 35 kernel calls to allow the rest of the operating system to do its work. These calls perform functions like hooking handlers to interrupts, moving data between address spaces, and installing new memory maps for newly created processes.

The process structure of MINIX 3 is divided into three parts above the kernel.

- The lowest layer contains device drivers.
- The middle layer includes servers.

Difference between Microkernel and Monolithic Kernel



Comparison Chart

BASIS FOR COMPARISON	MICROKERNEL	MONOLITHIC KERNEL
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BASIS FOR COMPARISON	MICROKERNEL	MONOLITHIC KERNEL
Basic	In microkernel user services and kernel, services are kept in separate address space.	In monolithic kernel, both user services and kernel services are kept in the same address space.
Size	Microkernel are smaller in size.	Monolithic kernel is larger than microkernel.
Execution	Slow execution.	Fast execution.
Extendible	The microkernel is easily extendible.	The monolithic kernel is hard to extend.
Security	If a service crashes, it does effect on working of microkernel.	If a service crashes, the whole system crashes in monolithic kernel.
Code	To write a microkernel, more code is required.	To write a monolithic kernel, less code is required.
Example	QNX, Symbian, L4Linux, Singularity, K42, Mac OS X, Integrity, PikeOS, HURD, Minix, and Coyotos.	Linux, BSDs (FreeBSD, OpenBSD, NetBSD), Microsoft Windows (95,98,Me), Solaris, OS-9, AIX, HP-UX, DOS, OpenVMS, XTS-400 etc.

III. Layered Systems Structure in Operating Systems

As the name suggests, this system works in layers. It was designed by E.W. Dijkstra in 1968, along with some help from his students. This system was first implemented in THE system built at the Technische Hogeschool Eindhoven in the Netherlands. The THE system was a simple batch system for a Dutch computer, the Extralogica X8.

Working

There are six layers in the system, each with different purposes.

Layer	Function
5	The operator
4	User Programs
3	Input/Output Management
2	Operator-process communication
1	Memory and drum management
0	Processor allocation and multiprogramming

Layer 0 – Processor Allocation and Multiprogramming – This layer deals with the allocation of processor, switching between the processes when interrupts occur or when the timers expire.

The sequential processes can be programmed individually without having to worry about other processes running on the processor. That is, layer 0 provides that basic multiprogramming of the CPU

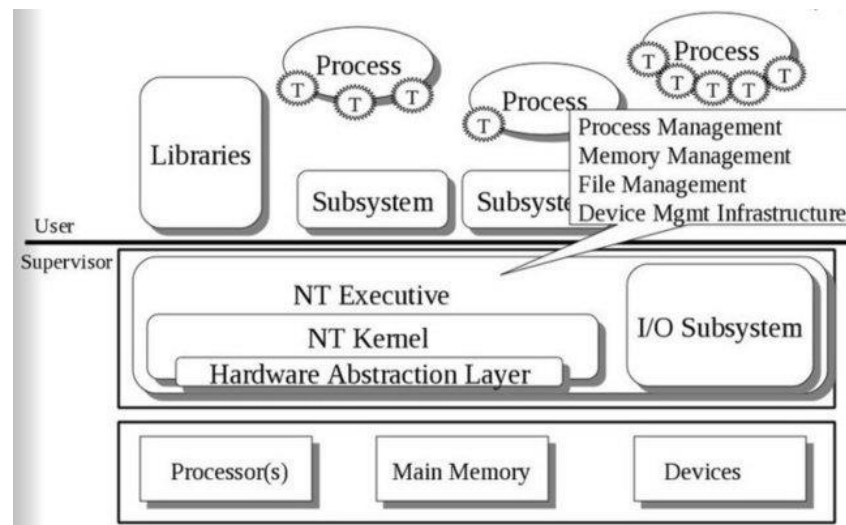
Layer 1 – Memory and Drum Management – This layer deals with allocating memory to the processes in the main memory. The drum is used to hold parts of the processes (pages) for which space couldn't be provided in the main memory. The processes don't have to worry if there is available memory or not as layer 1 software takes care of adding pages wherever necessary.

Layer 2 – Operator-Process communication – In this layer, each process communicates with the operator (user) through the console. Each process has its own operator console and can directly communicate with the operator.

Layer 3 – Input/Output Management – This layer handles and manages all the I/O devices, and it buffers the information streams that are made available to it. Each process can communicate directly with the abstract I/O devices with all of its properties.

Layer 4 – User Programs – The programs used by the user are operated in this layer, and they don't have to worry about I/O management, operator/processes communication, memory management, or the processor allocation.

Layer 5 – The Operator – The system operator process is located in the outer most layer.



IV. Client-Server Model in Operating Systems

The client-server model in an operating system is a variation of the microkernel system. The middle layer in the microkernel system is the one with servers. These servers provide some kind of service to clients. This makes up the client-server model.

Communication between clients and servers is obtained by message passing. To receive a service, one of the client processes constructs a message saying what it wants and sends it to the appropriate service. The service then does its work and sends back the answer.

If the clients and servers are on the same machine, then some optimizations are possible. But generally speaking, they are on different systems and are connected via a network link like LAN or WAN.

The best example of this model is you reading this article learning about it right now. You are the client, and you are requesting this page from whatever host this article has been uploaded to. The internet is basically the example since much of the web operates this way.

V. Virtual Machines in Operating System

Virtual Machine abstracts the hardware of our personal computer such as CPU, disk drives, memory, NIC (Network Interface Card) etc, into many different execution environments as per our requirements, hence giving us a feel that each execution environment is a single computer. For example, VirtualBox.

When we run different processes on an operating system, it creates an illusion that each process is running on a different processor having its own virtual memory, with the help of CPU scheduling and virtual-memory techniques. There are additional features of a process that cannot be provided by the hardware alone like system calls and a file system. The virtual machine approach does not provide these additional functionalities but it only provides an interface that is same as basic hardware. Each process is provided with a virtual copy of the underlying computer system.

We can create a virtual machine for several reasons, all of which are fundamentally related to the ability to share the same basic hardware yet can also support different execution environments, i.e., different operating systems simultaneously.

The main drawback with the virtual-machine approach involves disk systems. Let us suppose that the physical machine has only three disk drives but wants to support seven virtual machines. Obviously, it cannot allocate a disk drive to each virtual machine, because virtual-machine software itself will need substantial disk space to provide virtual memory and spooling. The solution is to provide virtual disks.

Users are thus given their own virtual machines. After which they can run any of the operating systems or software packages that are available on the underlying machine. The virtual-machine software is concerned with multi-programming multiple virtual machines onto a physical machine, but it does not need to consider any user-support software. This arrangement can provide a useful way to divide the problem of designing a multi-user interactive system, into two smaller pieces.

Advantages:

1. There are no protection problems because each virtual machine is completely isolated from all other virtual machines.

2. Virtual machine can provide an instruction set architecture that differs from real computers.
3. Easy maintenance, availability and convenient recovery.

Disadvantages:

1. When multiple virtual machines are simultaneously running on a host computer, one virtual machine can be affected by other running virtual machines, depending on the workload.
2. Virtual machines are not as efficient as a real one when accessing the hardware.

VI. Exokernels

Exokernels are a subset of virtual machines. In this, the disks are actually partitioned, and resources are allocated while setting it up. The different OS may be installed on different partitions. Beneath both, the partition is what we call the exokernel.